Jerry Zeng

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Education

University of Waterloo — Waterloo, ON
Candidate for Bachelor of Mathematics — *Sep 2025 – May 2030*

- **Awards:** President's Scholarship of Distinction
- **Relevant Courses:** Designing Functional Programs (Racket), Elementary Algorithm Design and Data Abstraction (C), Tools and Techniques for Software Development (testing/debugging, build automation, etc.)

Technical Skills

Programming Languages: Certified Associate Python Programmer (Python Institute), C, JavaScript, HTML/CSS
Frameworks & Libraries: React, FastAPI, Tailwind CSS, pandas, NumPy, Matplotlib
Tools & DevOps: Git, PyTest, Visual Studio Code, PyCharm, Vim, Figma, Linux, BASH, LaTeX
Spoken Languages : English *(Native)*, French *(DELF B2)*, Mandarin Chinese *(Conversational)*

Projects

WatTheDuck | *React.js, TypeScript, Django, REST, Gemini API, ElevenLabs API, ESP32* — May 2026

- Collaborated with 3 others over 72 hours on an AI-powered rubber duck debugging companion to assist developers in understanding their codebase and solving engineering problems, using the Socratic method
- Deployed a full-stack application integrating a React/TypeScript frontend, Django REST backend, conversational AI, and text-to-speech, enabling real-time verbal debugging sessions
- Engineered a physical hardware component using an ESP32 microcontroller that receives live signals triggered by a button and connected microphone

Winner at FraserHacks: **LogiCraft** | *Java, Spigot* — Top 3 of 31 – Nov 2024

- Developed Boolean logic education by developing an immersive Minecraft environment that utilizes Redstone circuitry and custom schematics to visualize AND, OR, and XOR gate operations
- Programmed a custom Java Spigot server plugin to automate interactive tutorials and track user progress, integrating the LightMagic library for real-time visual logic feedback, and iterated using automated testing

Carbon Footprint Calculator | *HTML, CSS, JS* — Dec 2022

- Architected a multi-step carbon footprint calculator for a sustainability initiative, estimating users' monthly CO2e emissions and translating results into a tree-offset equivalent to make environmental impact tangible
- Built with vanilla HTML, CSS, and JavaScript using Material Design Components for a responsive, framework-free UI, deployed on Vercel and embedded into a live sustainability web platform

Experience

University of Waterloo — Waterloo, ON
Web Accessibility Auditor — *May 2026 – Aug 2026*

- Audited 3 Professional Development courses (**30+ modules** and supplemental content) against **WCAG 2.1 AA** using D2L Accessibility Checker, axe, and browser developer tools
- Identified, logged and categorized **300+ recommendations** in Excel, tracking remediation status and separating actionable fixes from issues requiring elevated permissions
- Implemented fixes in D2L Brightspace, including transcript corrections and heading restructuring
- Consolidated findings into Word reports and presented problems to project supervisors during weekly reviews, documenting outstanding issues for future resolution

Helping Hands of Peel — Peel, ON
Vice-President — *Jul 2024 – Jul 2025*

- Managed a team of 50+ general members and executives, coordinating initiatives across committees
- Led the logistics and execution of a multi-week food drive, collecting **1,520 lbs** of donated food
- Planned, hosted, and oversaw 5 in-person and online events with 15+ attendees each, including volunteer initiatives at senior homes and a community park clean-up